

Project Proposal

Mapping the Genetic Landscape of Sleep and Circadian Biology Across Human Traits and Diseases

Central Hypothesis

Sleep and circadian biology are not merely behavioral traits. They represent a fundamental genetic and biological axis that connects multiple domains of human health, including brain disease, psychiatric disease, immune disease, metabolic disease, cardiovascular disease, cancer, and aging.

Genetic variants associated with sleep and circadian traits may contribute to disease susceptibility through shared pleiotropic loci, latent genetic factors, cell-type-specific regulatory programs, and conserved biological pathways.

Overall Goal

To construct a systematic Sleep/Circadian Genetic Atlas across human traits and diseases.

The project will answer four major questions:

1. Which diseases share genetic architecture with sleep and circadian traits?
 2. Which loci and variants drive this shared genetic architecture?
 3. Are there latent genetic factors linking sleep/circadian biology to disease domains?
 4. Which cell types and pathways mediate these shared genetic effects?
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Study Design Overview

The project will follow a structure similar to large-scale cross-disorder psychiatric genetics studies.

Sleep/Circadian GWAS

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Genome-wide genetic correlation

↓

Local genetic correlation

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Cross-trait pleiotropic loci

↓

Genomic SEM latent factors

↓

Fine-mapping and colocalization

↓

Cell-type and pathway annotation

↓
Prioritized mechanisms and disease domains

Dataset Collection

A. Sleep and Circadian GWAS Traits

Start with 6 core sleep/circadian traits:

1. Insomnia
2. Sleep duration
3. Short sleep
4. Long sleep
5. Chronotype / morningness-eveningness
6. Daytime sleepiness

Second-stage traits:

7. Napping
8. Snoring
9. Sleep apnea proxy
10. Sleep efficiency
11. Actigraphy-derived sleep duration
12. Activity rhythm / circadian amplitude
13. Sleep timing
14. REM/NREM-related phenotypes, if available

Priority sources:

- UK Biobank sleep GWAS
 - FinnGen
 - GWAS Catalog
 - OpenGWAS
 - SSGAC
 - Neale Lab UKB
 - PGC-related sleep/psychiatric datasets
 - Large published sleep GWAS summary statistics
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B. Disease and Trait Domains

Domain 1. Brain and Neurodegeneration

- Alzheimer's disease
- Parkinson's disease
- ALS
- Dementia
- Cognitive performance
- Educational attainment

Domain 2. Psychiatric Disorders

- Major depressive disorder
- Bipolar disorder
- Schizophrenia
- Autism spectrum disorder
- ADHD
- Anxiety
- PTSD
- OCD
- Substance use disorders

Domain 3. Immune and Inflammatory Diseases

- IBD
- Crohn's disease
- Ulcerative colitis
- Rheumatoid arthritis
- Multiple sclerosis
- Lupus
- Asthma
- Atopic dermatitis

Domain 4. Metabolic Traits

- BMI
- Obesity
- Type 2 diabetes
- Fasting glucose
- Fasting insulin
- Lipids
- NAFLD
- Waist-to-hip ratio

Domain 5. Cardiovascular Traits

- Coronary artery disease
- Hypertension
- Stroke
- Atrial fibrillation
- Heart failure
- Blood pressure traits

Domain 6. Aging and Longevity

- Longevity
- Parental lifespan
- Healthspan
- Frailty
- Epigenetic age acceleration
- Telomere length

Domain 7. Cancer

Start with major well-powered cancer GWAS:

- Breast cancer
 - Prostate cancer
 - Colorectal cancer
 - Lung cancer
 - Melanoma
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Phase 0. GWAS Curation and Harmonization

Goal

Create a clean, standardized GWAS summary statistics resource.

Input

For each trait:

- SNP ID
- Chromosome
- Position
- Effect allele
- Other allele
- Beta or odds ratio
- Standard error
- P-value
- Sample size
- Case/control number if applicable
- Ancestry
- Genome build

QC Steps

1. Keep European ancestry first.
2. Convert all files to hg19 or hg38 consistently.
3. Remove ambiguous SNPs: A/T and C/G if allele frequency is uncertain.
4. Keep HapMap3 SNPs for LDSC.
5. Remove SNPs with low imputation quality if INFO is available.
6. Remove duplicated SNPs.
7. Harmonize allele direction.
8. Estimate SNP heritability for each trait.
9. Exclude traits with low SNP heritability or unstable LDSC intercept.

Packages and Tools

R

- data.table
- tidyverse
- TwoSampleMR
- ieugwasr
- coloc
- susieR
- GenomicSEM
- lavaan
- pheatmap
- ComplexHeatmap
- ggplot2

Python / command line

- LDSC
- munge_sumstats.py
- PLINK
- bcftools
- liftOver
- pandas
- numpy
- scipy
- matplotlib

Reference files

- 1000 Genomes European LD reference
- HapMap3 SNP list
- LDSC baseline LD scores
- UK Biobank or 1000G LD panels

Output

A GWAS metadata table:

Trait	Domain	Source	PMID	Sample Size	Ancestry	Build	h2 SNP	File
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Phase 1. Genome-Wide Genetic Correlation

Goal

Identify global shared genetic architecture between sleep/circadian traits and diseases.

Main Method

LD Score Regression.

Tool

LDSC.

Analysis

For each sleep/circadian trait, run pairwise LDSC against all disease traits.

Example:

Insomnia × AD
Insomnia × MDD
Insomnia × IBD
Chronotype × SCZ
Chronotype × T2D
Sleep duration × BMI
Daytime sleepiness × CAD
Short sleep × longevity

Command Structure

```
munge_sumstats.py  
↓  
ldsc.py --rg trait1.sumstats.gz,trait2.sumstats.gz
```

Output Table

Sleep Trait	Disease Trait	Domain	rg	SE	P	FDR	h2_sleep	h2_disease
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Main Figure

A heatmap:

Sleep/circadian traits on rows
Disease traits on columns
Color = genetic correlation rg
Stars = FDR significance

Interpretation

This step identifies which disease domains are most genetically connected to sleep/circadian biology.

Expected patterns:

- Insomnia: psychiatric and neurodegenerative traits
 - Chronotype: psychiatric, metabolic, behavioral traits
 - Sleep duration: metabolic and aging traits
 - Daytime sleepiness: cardiometabolic and neurodegenerative traits
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Phase 2. Local Genetic Correlation

Goal

Genome-wide r_g may miss regional effects. Local genetic correlation identifies genomic regions where sleep and disease share genetic effects.

Methods

Use:

- LAVA
- SUPERGENOVA
- ρ -HESS

Recommended First Choice

LAVA.

Input

- Harmonized GWAS summary statistics
- 1000 Genomes European LD reference
- Predefined independent LD blocks

Analysis

Run LAVA only for prioritized trait pairs from Phase 1.

Priority criteria:

1. FDR-significant LDSC r_g
2. Large absolute r_g
3. Biological relevance
4. Well-powered GWAS
5. Relevant to your downstream single-cell data

Example top pairs:

- Insomnia × MDD
- Insomnia × AD
- Chronotype × SCZ
- Sleep duration × BMI
- Daytime sleepiness × CAD
- Sleep duration × longevity
- Insomnia × IBD

Output Table

Regio	Sleep Trait	Disease	Local r_g	P	FDR	Candidate
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Main Figure

Manhattan-style plot of local genetic correlation regions.

Interpretation

This step tells us where shared genetic effects occur.

Phase 3. Cross-Trait Pleiotropic Locus Discovery

Goal

Identify variants and loci associated with both sleep/circadian traits and diseases.

Methods

Use at least two complementary methods:

1. PLACO
2. Conjunction FDR
3. CPASSOC
4. ASSET
5. MTAG, only for closely related traits

Recommended First Pass

Use PLACO and conjunction FDR.

Why

PLACO is useful for identifying pleiotropic variants when both traits may have effects.
Conjunction FDR is useful for ranking variants jointly associated with two traits.

Input

- Harmonized summary statistics
- Matched SNP sets
- LD-clumped loci

Output Table

SNP	Chr	Pos	Sleep Trait	Disease Trait	Beta Sleep	P Sleep	Beta Disease	P Disease	PLACOP	ConjFDR	Direction	Nearest Gene
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Direction Categories

For each shared SNP:

- Same direction: risk-increasing for both traits
- Opposite direction: increases sleep trait but decreases disease risk
- Complex/unclear

Main Figure

Shared loci circus plot or genome-wide Manhattan plot.

Interpretation

This gives the concrete shared loci atlas.

Phase 4. Genomic SEM and Latent Factor Modeling

Goal

Identify latent genetic factors linking sleep/circadian traits with disease domains.

This is the key step inspired by the psychiatric cross-disorder papers.

Main Package

GenomicSEM in R.

Input

- LDSC genetic covariance matrix
- SNP-level summary statistics for multivariate GWAS
- Well-powered traits only

Step 4.1. Build Genetic Correlation Matrix

Include:

Sleep/circadian traits

- psychiatric diseases
- neurodegenerative diseases
- immune diseases
- metabolic traits
- cardiovascular traits
- aging traits

Step 4.2. Exploratory Factor Analysis

Use:

- factanal
- psych package
- GenomicSEM
- lavaan

Goal:

Identify clusters such as:

Sleep Disturbance Factor

- Insomnia
- Short sleep
- Daytime sleepiness

Circadian Timing Factor

- Chronotype
- Sleep timing
- Activity rhythm

Psychiatric/Internalizing Factor

- MDD
- Anxiety
- PTSD

Severe Psychiatric Factor

- SCZ
- BIP

Immune Factor

- IBD
- RA
- MS
- Lupus

Metabolic Factor

- BMI
- T2D

- NAFLD
- Lipids

Aging Factor

- Longevity
- Frailty
- Healthspan

Step 4.3. Confirmatory Factor Analysis

Fit candidate models:

Model 1: Sleep traits independent from disease factors

Model 2: Sleep disturbance loads with psychiatric/internalizing factor

Model 3: Circadian timing connects psychiatric and metabolic factors

Model 4: Sleep/circadian factors act as central upstream genetic axes

Evaluate model fit:

- CFI
- SRMR
- AIC
- BIC
- χ^2
- factor loadings

Step 4.4. Multivariate GWAS of Latent Factors

Use GenomicSEM userGWAS.

Output:

SNP associations with:

- Sleep Disturbance Factor
- Circadian Factor
- Sleep–Psychiatric Shared Factor
- Sleep–Metabolic Shared Factor
- Sleep–Aging Shared Factor

Output Table

Factor	SNP	Chr	Pos	Beta	SE	P	Nearest Gene	Novel/known
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Main Figure

Genetic factor model diagram.

Interpretation

This step asks:

Does sleep/circadian biology form a distinct genetic factor, or does it integrate into broader disease factors?

Phase 5. Disorder-Specific vs Shared Effects

Goal

Separate shared sleep/disease genetic effects from trait-specific effects.

Methods

Use:

- GenomicSEM common factor GWAS
- GenomicSEM Q_SNP heterogeneity test
- GWAS-by-subtraction
- CC-GWAS if appropriate

Key Questions

1. Which loci are shared across sleep and many diseases?
2. Which loci are specific to sleep traits?
3. Which loci are specific to individual disease domains?
4. Which loci show heterogeneous effects across diseases?

Output

SNP	Shared Factor P	Q_SNP P	Sleep-specific c	Disease-specific c	Interpretation
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Interpretation

This is analogous to separating pleiotropic and disorder-specific loci in psychiatric genetics.

Phase 6. Fine-Mapping and Colocalization

Goal

Determine whether shared loci are driven by the same causal variant.

Fine-Mapping Tools

- susieR
- FINEMAP
- PolyFun + SuSiE
- DAP-G

Colocalization Tools

- coloc
- eCAVIAR
- HyPrColoc
- moloc
- SMR/HEIDI

Recommended Pipeline

For each high-priority locus:

1. Define ± 500 kb or LD-based locus.
2. Fine-map sleep trait.
3. Fine-map disease trait.
4. Compare credible sets.
5. Run coloc.
6. Prioritize $PP.H4 > 0.8$.
7. Annotate candidate causal variants.

Output

Locu s	Sleep Trait	Disease Trait	Lead SNP	Credible Set	PP.H4	Candidate Gene	Evidenc e
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Interpretation

Shared locus does not equal shared mechanism.
Colocalization is required to identify true shared causal signals.

Phase 7. TWAS, PWAS, and Molecular Mediation

Goal

Link shared genetic risk to genes, proteins, and molecular mechanisms.

TWAS Tools

- FUSION
- S-PrediXcan
- S-MultiXcan
- UTMOST

eQTL/sQTL Resources

- GTEx
- PsychENCODE
- eQTLGen
- ROSMAP
- CommonMind
- Brain xQTL
- immune eQTL datasets

PWAS / Protein Tools

- SMR
- PWAS
- pQTL colocalization
- Open Targets Genetics

Output

Gene	Tissue	Trait Pair	TWAS Z	P	Coloc PP	Direction
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Interpretation

This step prioritizes genes whose genetically regulated expression may mediate sleep–disease shared risk.

Phase 8. Cell-Type-Specific Functional Genomics

Goal

Identify the cellular contexts where shared sleep/disease variants act.

This is where the project becomes mechanistic and distinct from pure GWAS studies.

Brain Cell Types

Use your AD and aging brain multiome atlas.

Cell types:

- Excitatory neurons
- Inhibitory neurons
- Astrocytes
- Microglia
- Oligodendrocytes
- OPCs
- Endothelial cells

- Pericytes

Immune Cell Types

Use public immune single-cell and ATAC datasets.

Cell types:

- T cells
- B cells
- Monocytes
- Macrophages
- Dendritic cells
- NK cells

Metabolic / Vascular Tissues

Use public tissue annotations:

- Liver
- Adipose
- Pancreatic islets
- Endothelium
- Vascular smooth muscle

Methods

Partitioned Heritability

- S-LDSC
- LDSC-SEG

Gene-Based / Gene-Set

- MAGMA
- FUMA
- DEPICT

Variant-to-Enhancer Mapping

- GARFIELD
- GREGOR
- bedtools overlap
- TORUS

Single-cell Regulatory Annotation

- ArchR
- Signac
- Cicero
- ABC model
- GREAT
- chromVAR

Specific Questions

1. Are insomnia–AD loci enriched in microglial enhancers?
2. Are chronotype–SCZ loci enriched in excitatory neuron enhancers?
3. Are sleep–IBD loci enriched in T cell or macrophage regulatory elements?
4. Are sleep–metabolic loci enriched in liver, adipose, or hypothalamic regulatory regions?
5. Are sleep–aging loci enriched in senescence or mitochondrial pathways?

Output

Trait Pair Cell Type Annotation Enrichment P FDR

Main Figure

Cell-type enrichment heatmap:

Rows = sleep/disease shared loci

Columns = cell types

Color = enrichment Z score

Phase 9. Pathway and Network Analysis

Goal

Convert loci and cell-type enrichments into biological mechanisms.

Tools

- MAGMA
- FUMA
- g:Profiler
- Enrichr
- GREAT
- MSigDB
- ReactomePA
- clusterProfiler
- WGCNA
- igraph
- Cytoscape

Pathway Categories

Circadian Clock

- CLOCK
- ARNTL / BMAL1
- PER1/2/3

- CRY1/2
- NR1D1
- RORA

Neurobiology

- Synaptic plasticity
- Neuronal excitability
- Neurotransmission
- Myelination

Immune Biology

- NF- κ B
- Interferon signaling
- Cytokine signaling
- Microglial activation

Metabolism

- AMPK
- mTOR
- insulin signaling
- mitochondrial function

Aging

- DNA damage repair
- senescence
- proteostasis
- epigenomic erosion
- inflammation

Output

A Shared Pathway Atlas.

Phase 10. Causal Inference

Goal

Test possible directionality.

Methods

- TwoSampleMR
- MR-Egger
- Weighted median
- MR-PRESSO

- CAUSE
- LHC-MR
- bidirectional MR

Analyses

Sleep/circadian trait → disease

Disease → sleep/circadian trait

Examples:

- Insomnia → AD
- AD → insomnia
- Short sleep → T2D
- T2D → short sleep
- Chronotype → depression
- Depression → chronotype
- Daytime sleepiness → CAD
- CAD → daytime sleepiness

Interpretation Rules

Do not overclaim causality.

Prioritize causal claims only when:

1. IVs are strong.
 2. MR methods agree.
 3. No strong horizontal pleiotropy.
 4. Colocalization supports shared variants.
 5. Biological annotation is consistent.
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Practical MVP Version

Do not start with 200 traits.

Start with:

6 Sleep/Circadian Traits

1. Insomnia
2. Sleep duration
3. Short sleep
4. Long sleep
5. Chronotype
6. Daytime sleepiness

40 Disease Traits

Brain / Psychiatric

- AD
- PD
- ALS
- MDD
- BIP
- SCZ
- ASD
- ADHD
- Anxiety
- PTSD

Immune

- IBD
- Crohn's disease
- Ulcerative colitis
- RA
- MS
- Lupus

Metabolic

- BMI
- Obesity
- T2D
- fasting glucose
- fasting insulin
- HDL
- LDL
- triglycerides
- NAFLD

Cardiovascular

- CAD
- hypertension
- stroke
- atrial fibrillation
- heart failure

Aging

- longevity
- parental lifespan
- healthspan
- frailty
- telomere length

Cancer

- breast cancer
- prostate cancer
- colorectal cancer

- lung cancer
- melanoma

This gives approximately 240 sleep/circadian × disease comparisons, which is manageable.

Recommended Manuscript Structure

Figure 1

Study design and GWAS dataset overview.

Figure 2

Genome-wide genetic correlation heatmap.

Figure 3

Latent genetic factor model using Genomic SEM.

Figure 4

Local genetic correlation and shared loci atlas.

Figure 5

Fine-mapping and colocalization of top loci.

Figure 6

Cell-type enrichment across brain, immune, metabolic, and vascular tissues.

Figure 7

Integrated model of sleep/circadian biology as a genetic axis of human disease.

Recommended Timeline

Month 1

Collect GWAS summary statistics.
Build metadata table.

Harmonize files.
Run LDSC SNP heritability QC.

Month 2

Run genome-wide LDSC genetic correlation.
Generate sleep × disease rg matrix.
Prioritize top trait pairs.

Month 3

Run LAVA for top trait pairs.
Run PLACO/conjunction FDR.
Identify shared loci.

Month 4

Run Genomic SEM.
Identify latent sleep/circadian-disease factors.
Run factor GWAS.

Month 5

Fine-map and colocalize top loci.
Run TWAS and eQTL colocalization.

Month 6

Run cell-type enrichment using brain, immune, metabolic, and vascular annotations.
Generate figures.

Month 7–8

Pathway analysis.
MR.
Draft manuscript or proposal.

Minimal Computing Environment

R Packages

- data.table
- tidyverse
- GenomicSEM
- lavaan
- psych

- TwoSampleMR
- ieugwasr
- coloc
- susieR
- coloc
- MAGMA.Celltyping
- clusterProfiler
- ReactomePA
- ComplexHeatmap
- pheatmap
- ggplot2

Python / Command Line

- LDSC
- munge_sumstats.py
- PLINK
- bcftools
- bedtools
- liftOver
- pandas
- numpy
- scipy
- matplotlib
- snakemake or nextflow

Genetics Tools

- LDSC
- LAVA
- SUPERGENOVA
- p-HESS
- PLACO
- CPASSOC
- ASSET
- MTAG
- GenomicSEM
- FINEMAP
- SuSiE
- coloc
- eCAVIAR
- SMR
- FUSION
- S-PrediXcan
- MAGMA
- FUMA
- GARFIELD
- GREGOR

Single-Cell Tools

- ArchR
- Signac
- Seurat

- Scanpy
 - Cicero
 - chromVAR
 - GREAT
 - bedtools
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Key Innovation

The innovation is not simply doing LDSC between sleep and disease.

The innovation is building a systematic sleep/circadian genetic atlas across the human phenome and then interpreting shared loci through cell-type-specific regulatory genomics.

The central model is:

Sleep/Circadian Genetics

↓

Shared Pleiotropic Loci

↓

Latent Disease Factors

↓

Cell-Type-Specific Regulatory Elements

↓

Disease-Specific and Shared Biological Pathways

↓

<https://pan.ukbb.broadinstitute.org/>

The UK Biobank is a collection of a half million individuals with paired genetic and phenotype information that has been enormously valuable in studies of genetic etiology for common diseases and traits. However, most genome-wide analyses of this dataset use only the European ancestry individuals. Analyzing a more inclusive and diverse dataset increases power and improves the potential for discovery. Here, we present a multi-ancestry analysis of 7,228 phenotypes, across 6 continental ancestry groups, for a total of 16,131 genome-wide association studies. We release these summary statistics freely to the community ahead of publication.

We acknowledge that genetic studies in diverse-ancestry individuals are culturally sensitive. We therefore regularly engaged with and sought feedback from a local affinity group for the advancement of ethnic minorities, several scholars and clinicians with experience working with minority groups, and a bioethics expert. They helped direct and contextualize this research, discussed actions to maximize benefits and minimize risks, and provided feedback on [an FAQ page](#) describing our study, its risks, benefits, and limitations. We intend for this resource to promote more inclusive research practices, accelerate novel scientific discoveries that are more generalizable, and improve the health of all people equitably.

Together, these efforts vastly extend the available resources for interpretation of disease-associated variants across diverse populations and highlight the importance of increasing diversity in genetic studies.

